Prasanna Reddy Pulakurthi

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Education

Rochester Institute of Technology, NY — *Aug 2019 - Dec 2025*
Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering | GPA: 3.93/4.0 — Rochester, NY, USA

Rochester Institute of Technology, NY — *Aug 2017 - Jul 2019*
Master of Science (MS) in Electrical Engineering | **GPA: 4.0/4.0** — Rochester, NY, USA

Technical Skills

- **LANGUAGES & SOFTWARE TOOLS:** Python, C, HTML/CSS, MATLAB, Linux, Unix, ROS, Gazebo, Docker, LaTeX, Git, SQL, Blender
- **AI/ML:** PyTorch, TensorFlow, TensorBoard, Keras, CUDA, GPU, PyTorch DDP, TensorRT, ONNX, NumPy, Scikit-Learn, data structures, OpenCV, mmCV, Hugging Face, Transformers, LLM (Large Language Models), NLP (Natural Language Processing), MMAction2, Path Planning, ORB-SLAM, CLIP, OpenAI, BLIP-2, LLaVA, Qwen-VL, MLLM

Professional Experience

AI/ML Researcher (PhD) – Rochester Institute of Technology — **Aug 2024 – Dec 2025**

- **Multimodal Reasoning (X-CoT, EMNLP 2025):** Designed an LLM-based chain-of-thought retrieval system improving R@1 by **+5.6%** on MSVD and producing explainable text-video retrieval.
- Developed two novel **Data Augmentation** techniques (Dual-region and Shuffle PatchMix) to improve model robustness and generalization. Demonstrated improvements in classification performance on domain adaptation datasets such as PACS by **+7.3 pp** and person re-identification datasets such as Market-1501 by up to **+6.91 pp**.

Perception Engineer Intern – DEVCOM Army Research Laboratory (via MBO Partners) — **May 2023 – Aug 2023**

- Advanced Generative Models: Developed a GAN variant combining neural-architecture search, PMish activation, bounded MMD repulsive loss, and adaptive rank decomposition, achieving a reduction of FID scores by **34%** (CIFAR-10) and **28%** (STL-10) while compressing the generator **4–10×**, enabling real-time edge deployment.
- Data Generation: Generated new synthetic gesture video data for Human Action Recognition using GANs and significantly lifted human-action recognition accuracy across ground **(+4.4 pp)** & aerial **(+7.7 pp)** views.

Perception Engineer Intern – DEVCOM Army Research Laboratory (via MBO Partners) — **May 2022 – Aug 2022**

- Developed domain adaptation techniques to improve model robustness across different digit domains, lifting cross-domain accuracy by up to **+2.56** percentage points (pp) on difficult shifts.
- Fast Classification Training: Formulated a fast, globally-optimal Iterative Maximum Likelihood Classifier by deriving a fixed-point solver from a regularized ML objective, achieving **~3× faster** convergence compared to gradient descent.

Engineering Intern – Qualcomm (VR Team) — **May 2021 – Aug 2021**

- Developed a temperature-compensation algorithm enhancing the accuracy of 6DOF 4Tx ultrasound VR controller.
- Implemented the algorithm in C, improving the root mean squared error (RMSE) by **+11.29 pp** on synthetic log data.

Engineering Intern – Microsoft (Iris Recognition, HoloLens-2 AR Team) — **May 2020 – Aug 2020**

- Optimized Gabor filter frequencies in Iris Recognition pipeline, significantly enhancing signature uniqueness.
- Developed GAN-based Super Resolution to enhance low-res Iris images while retaining key Gabor frequencies.

Research Assistant – Oak Ridge National Laboratory — **May 2018 – Jul 2019**

- Designed and executed an unmanned aerial system (UAS) data collection to capture aerial imagery of a building casting shadows on 14 differently coloured panels, both in and out of shadow.
- Developed six algorithms for shadow detection in aerial images using colour transforms and ML techniques, leveraging collected data. Improved upon traditional approaches by **+1.51%** in accuracy and **+2.4%** in F-score.

Select Publications

- Prasanna Reddy Pulakurthi et al. "X-CoT: Explainable Text-to-Video Retrieval via LLM based Chain-of-Thought Reasoning," **EMNLP, 2025.**
- Prasanna Reddy Pulakurthi et al. "Shuffle PatchMix Augmentation with Confidence Margin Weighted Pseudo Labels for Enhanced Source-Free Domain Adaptation," **IEEE ICIP, 2025.**
- Prasanna Reddy Pulakurthi et al. "Effective dual-region augmentation for reduced reliance on large amounts of labeled data," Synthetic Data for Artificial Intelligence and ML: Tools, Techniques, and Applications III. SPIE, 2025.
- Prasanna Reddy Pulakurthi et al. "Enhancing GANs with MMD Neural Architecture Search, PMish Activation Function and Adaptive Rank Decomposition," **IEEE Access, 2024**.
- Prasanna Reddy Pulakurthi et al. "Enhancing human action recognition with GAN-based data augmentation," Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications II. SPIE, 2024.
- Prasanna Reddy Pulakurthi et al. "Enhancing GAN Performance through Neural Architecture Search and Tensor Decomposition," **IEEE ICASSP, 2024.**

Projects

Machine Learning and Generative AI Projects

- Utilized GANs to generate a synthetic video dataset for Human Action Recognition.
- Applied GANs to unsupervised/source-free domain adaptation and image-to-image translation tasks.
- Employed Stable Diffusion with Control-Net to create a gesture recognition dataset with multiple views.
- Developed Invertible Neural Networks (INNs) to perform super-resolution and image denoising.

Vision Captioning Assistant (V-CAT)

- Built a full-stack, camera-driven "smart-glasses" assistant that streams live video to vision-language models and returns real-time scene captions plus synthesized speech for hands-free accessibility.

Signmentor AI (ASL Sign Tutoring)

- Built a full-stack website for ASL sign tutor that accurately identifies handshapes in the ASL alphabet (A to Z) by pairing your camera feed with cutting-edge Multimodal LLM for fast, reliable classification and learner feedback.

Comparison of Simultaneous Localization and Mapping (SLAM) Algorithms in ROS Environment

- Simulated and benchmarked Gmapping (2D laser-based), Octomap (probabilistic 3-D occupancy-grid), and RTAB-Map (RGB-D dense mapping) in a Gazebo/ROS indoor environment using custom and stock robots outfitted with LiDAR and RGB-D sensors.
- Evaluated map quality, runtime, and sensor cost, recommending Gmapping for lightweight real-time 2-D mapping, Octomap for memory-efficient 3-D obstacle maps, and RTAB-Map for high-fidelity textured reconstructions.

Image and Signal Processing Projects

- Implemented several image and signal processing algorithms, such as image resizing, noise filtering, image enhancement, image segmentation, camera calibration, stereo reconstruction, affine, and projective transforms.

Real-Time Attendance System using Image Processing

- Developed a real-time automatic attendance system using OpenCV for face detection, Python-implemented Binarized Statistical Image Features for face recognition, and a Kivy-based app interfacing via sockets and PHP with an Apache-hosted GUI displaying attendance data stored in MySQL.

Automatic License Plate Detection and Recognition

- Built an automatic number plate detection and recognition system using edge and histogram-based image processing methods for detection and neural networks for recognition, coded in MATLAB.

Coursework

Digital Signal Processing, Digital Image Processing, Multi-view Imaging, Artificial Intelligence Explorations, Pattern Recognition, Information Theory Coding, Image and Video Compression, Advanced Engineering Mathematics